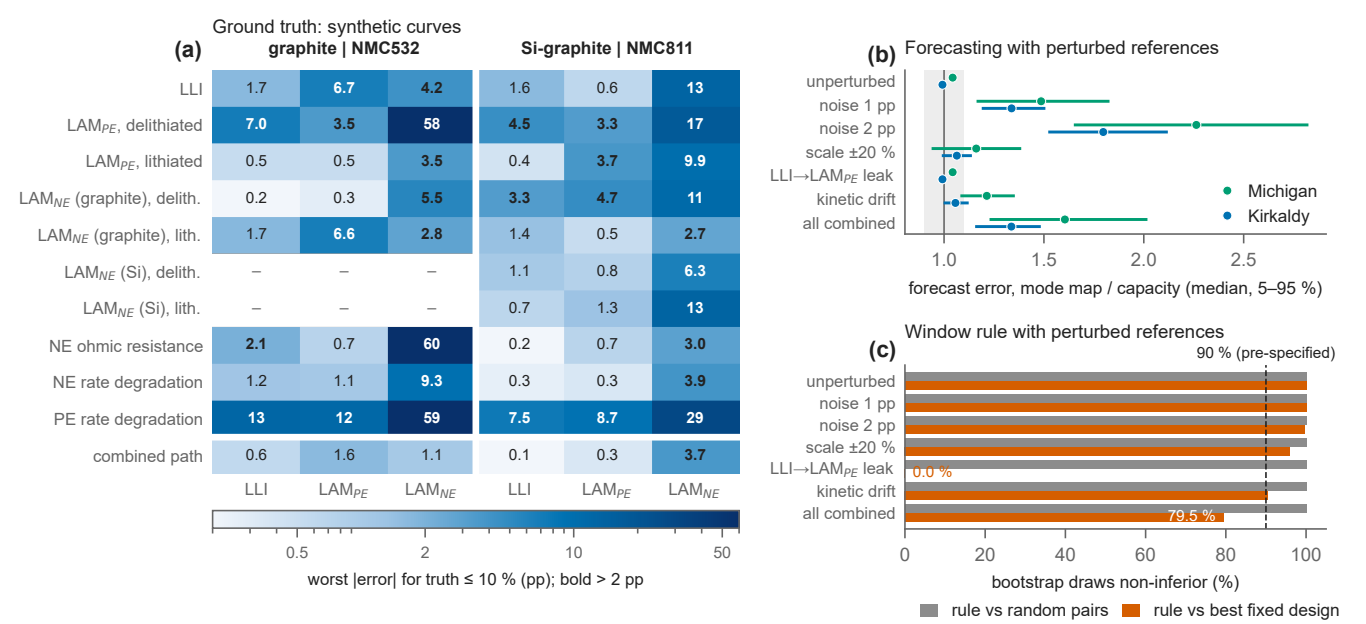

(a)
Ground truth: synthetic curves
graphite | NMC532
Si-graphite | NMC811
LLI
LAM$_{PE}$, delithiated
LAM$_{PE}$, lithiated
LAM$_{NE}$ (graphite), delith.
LAM$_{NE}$ (graphite), lith.
LAM$_{NE}$ (Si), delith.
LAM$_{NE}$ (Si), lith.
NE ohmic resistance
NE rate degradation
PE rate degradation
combined path
1.7 6.7 4.2 1.6 0.6 13
7.0 3.5 58 4.5 3.3 17
0.5 0.5 3.5 0.4 3.7 9.9
0.2 0.3 5.5 3.3 4.7 11
1.7 6.6 2.8 1.4 0.5 2.7
– – – 1.1 0.8 6.3
– – – 0.7 1.3 13
2.1 0.7 60 0.2 0.7 3.0
1.2 1.1 9.3 0.3 0.3 3.9
13 12 59 7.5 8.7 29
0.6 1.6 1.1 0.1 0.3 3.7
LLI
LAM$_{PE}$
LAM$_{NE}$
LLI
LAM$_{PE}$
LAM$_{NE}$
0.5
2
10
50
worst |error| for truth ≤ 10 % (pp); bold > 2 pp
(b)
Forecasting with perturbed references
unperturbed
noise 1 pp
noise 2 pp
scale ±20 %
LLI→LAM$_{PE}$ leak
kinetic drift
all combined
Michigan
Kirkaldy
1.0
1.5
2.0
2.5
forecast error, mode map / capacity (median, 5–95 %)
(c)
Window rule with perturbed references
90 % (pre-specified)
unperturbed
noise 1 pp
noise 2 pp
scale ±20 %
LLI→LAM$_{PE}$ leak
kinetic drift
all combined
0.0 %
79.5 %
0
20
40
60
80
100
bootstrap draws non-inferior (%)
rule vs random pairs
rule vs best fixed design